Ethan Pichon

Washington, DC | epichon@caltech.edu | 202-607-8568

https://www.linkedin.com/in/ethan-pichon-93934a243/ | github.com/epichon1

Education

California Institute of Technology, BS in Mechanical Engineering with Robotics Minor — Sept 2023 – Jun 2027

- GPA: 4.0/4.0
- **Coursework:** Mobile Robots, Experimental Robotics, Fluid Mechanics, Mechanics, Robotics Planning, Design and Fabrication, Thermal Science
- **Club Roles:** NSBE Co-President, BSU Secretary, CAOS Rover Team, PARSEC Controls Team, Chess Club President

Skills

Mechanical: Solidworks, ANSYS, Fabrication, Machining(Lathe, Mill, Band Saw, WaterJet), Mechanical Design, 3D-Printing

Software: ROS2, Github, Jira, C++, Simulink, Python, Java, MATLAB, Miscrosoft Office, Rasberry Pi

Robotics: Robotic planning, Inverse/Forward Kinematics, Controls, RVIZ, Gazebo

Experience

Robotics Teaching Assistant, Caltech – Pasadena, CA — Mar 2026 – Jun 2026

- Designed and fabricated mobile robot kits using laser-cutting techniques for a class of 14 students.
- Developed ROS2 networked python script to generate waypoints with cost estimation for competition
- Conducted research on object distinction using LiDAR scanning, improving obstacle avoidance accuracy by 65%.

Robotics Research Assistant, UMD Space Systems Lab – College Park, MD — Jul 2025 – Oct 2025

- Implemented automation planning algorithms within ROS2 for a prototype human-space colony rover.
- Derived inverse kinematic equations tailored to specific wheel constraints to optimize rover maneuverability and stability.
- Evaluated multiple planning algorithms (EST, RRT) and post-processing methods for human-interpretable path planning by surveying human trajectory predictions.

Emergency Engineer Consultant, SEDC – Washington, DC — Apr 2023 – Sept 2023

- Authored comprehensive building safety reports and provided actionable recommendations to ensure compliance with municipal fire safety regulations.
- Evaluated and pitched alternative CAD software solutions to management, projecting a 30% increase in building visualization efficiency.

CAD Design Intern, Michael Graves – Washington, DC — Jan 2023 – Jun 2023

- Created CAD visualizations in Solidworks for Architecture Showcases
- 3D Printed Building Landscapes for Architecture Presentations

Co-Owner/Graphic Design Lead, POE Maps – Washington, DC — Jun 2020 – Present

- Spearheaded graphic design for two published games featured on the Mojang homepage, generating over 5 million global views.
- Organized and executed community tournaments, successfully cultivating and managing an active player base of 100+ members.

Research

Reinforcement Learning Benchmarking, Robert T. Herzog SURF Fellow, Riviere Robot Lab – NYU Tandon — Jun 2026 - Sep 2026

- Created 3 OOS based space manipulation tasks in a mujoco warp gym env python sim and constructed in real world
- Trained Benchmark RL and Sample Based Algorithms (PPO,MPPI,CEM) in sim and deployed in real
- Analyzed performance against novel RL algorithm to submit to ICRA

Temporal Robotics Planning — Jan 2025 - Mar 2025

- Engineered a Python-based simulation of two robots (hider and seeker) navigating a continuous space across multiple map environments.
- Implemented temporal EST and RRT planning to evaluate evasion and pursuit tactics.
- Conducted a comparative analysis of planners, determining that RRT yielded a 22% higher success rate for the hider by providing more direct escape paths.

DNA Storage — Jan 2024 - Mar 2024

- Developed a Python-based computational model of a DNA storage system to calculate estimated data retrieval and write times.
- Leveraged parallel processing and oligonucleotides to significantly reduce runtime, achieving a processing speed of 10 nucleotides per second.

Projects

Capstone Design Robot — Sept 2025 - Mar 2026

- Executed the full hardware lifecycle (PDR, CDR, BOM) for a custom robotic system, managing timeline delivery via Gantt charts.
- Developed a C++ motor controller utilizing PID control and an ESP-32/Roboclaw stack to enable full autonomy.
- Designed a custom agitation and funneling system capable of intake and expulsion within a 3-inch diameter without jamming.
- Machined and assembled a highly robust 40 lb robot capable of scaling a 37-degree incline.

SLAM Robot — Apr 2025 - Jun 2025

- Constructed a ROS2-powered SLAM robot utilizing an RRT algorithm and least-squares fit for real-time localization.
- Derived inverse kinematics and geometric identities to ensure precise mechanical driving and accurate spatial mapping.
- Implemented dynamic coverage planning and obstacle avoidance for real-world navigation applications.

Intelligent Line Follower — Apr 2024 - Jun 2024

- Constructed a line-following robot driven by an A* algorithm, integrating IR and ultrasound sensors for environmental feedback.
- Implemented multithreading architecture and a concise UI, allowing users to dynamically alter the robot's objective in real-time.

Tracking Camera — Sept 2024 - Dec 2024

- Engineered a motorized tracking camera system, fully CADing and assembling the custom housing and motor mounts.
- Optimized visual tracking by applying advanced noise reduction techniques, including HSV constriction, erode-dilate filtering, and shape recognition.

Planetary Transmission — Apr 2025 - Jun 2025

- Delivered comprehensive PDR and CDR documentation, including a full BOM, CAD models, and structural research analysis.
- Precision-machined and successfully assembled a fully functional planetary transmission system.